The Journal of Child Psychology and Psychiatry

Journal of Child Psychology and Psychiatry 67:5 (2026), pp 605–608 doi:10.1111/jcpp.70150

Editorial: ADHD persistence – the interplay of genes, socioeconomic context, and symptom domains over development

ADHD persistence as a developmental question

The persistence of attention-deficit/hyperactivity disorder (ADHD) symptoms into adulthood is common, associated with adverse outcomes in multiple domains including education, occupation, and comorbid mental health disorders (Caye et al., 2016; Faraone et al., 2024). A binary label of persistence, however, obscures the reality that for many individuals ADHD symptoms vary considerably across time (Sibley et al., 2022). Inattention and hyperactivity/impulsivity show different developmental trajectories and are differentially associated with long-term outcomes (Faraone et al., 2024; Pingault et al., 2015; Sonuga-Barke et al., 2023). For example, hyperactivity/impulsivity typically declines from childhood into adulthood and is more strongly linked to externalizing and risk-taking outcomes, whereas inattention is relatively more stable and more strongly associated with persistent academic and occupational impairment (Garner et al., 2013). Current evidence recognizes that genetic contributions to ADHD symptoms vary by age and symptom domain (Pingault et al., 2015), but the importance of environmental factors such as socioeconomic context for symptom persistence remains underexamined (Rakesh, Lee, Gaikwad, & McLaughlin, 2025). We present evidence supporting a unified framework for conceptualizing ADHD symptom persistence as a dynamic developmental process shaped by three components: (1) stable genetic liability, (2) time-varying gene–environment interplay, and (3) independent environmental effects by symptom domain. We focus on socioeconomic status (SES) as a case study, to guide future research applying genetic and environmental data to predict risk for ADHD symptom persistence over the long term.

Conceptual foundations: Heritability and environmental context

ADHD is a highly heritable condition, with twin studies reporting mean heritability estimates of around 70–80% (Faraone et al., 2024). Genome-wide genetic risk for ADHD, reflected in polygenic risk scores, is also consistently associated with ADHD diagnosis. However, such polygenic scores do not necessarily differ between those with remitted versus persistent ADHD in the long term (Agnew-Blais et al., 2021). This suggests that genetic liability may be a stronger predictor of stable, early-emerging symptom levels than of developmental change (Pingault et al., 2015).

There are two possible nonmutually exclusive explanations for this. First, different genes and genetic mechanisms may relate to baseline symptom levels and changes over time, which are not well-captured by current polygenic risk scores derived from case–control genome-wide association studies, predominantly conducted in cross-sectional adult samples. Without genome-wide studies explicitly modelling symptom changes, the distinction between stable versus changing genetic influences remains difficult to test (Pingault et al., 2022). Addressing this may require either large, pooled datasets spanning multiple age groups or – more ambitiously – large-scale longitudinal genetic studies. Second, while genetics may be the predominant influence on childhood ADHD presence, environmental factors may exert a unique and strong influence on its developmental trajectory. For example, SES represents one of most important contextual influences associated with ADHD, linked to both higher rates of prevalence (Rowland et al., 2018) and persistence (Kim, King, & Jennings, 2019). Multiple mechanisms may link environmental factors and ADHD, including (1) gene–environment correlation with genetic factors influencing both SES and ADHD (Sonuga-Barke et al., 2023), (2) gene–environment interaction where SES moderates genetic liability (e.g., through differential exposure to stress), or (3) independent SES effects on longitudinal brain and cognitive development (Rakesh et al., 2025). Thus, symptom changes may reflect a reweighting of contributions over time from genetic liability, environmental effects, and environmental modulation of genetic liability, which have somewhat distinct impacts on inattention and hyperactivity/impulsivity domains. Below, we briefly introduce evidence and hypotheses for these ideas.

ADHD symptom persistence as a dynamic developmental process shaped by three components: (1) stable genetic liability, (2) time-varying gene–environment interplay, and (3) independent environmental effects by symptom domain.

Conflict of interest statement: No conflicts declared.

© 2026 Association for Child and Adolescent Mental Health.

Process I. Genetic liability

Progress in unravelling the genetic architecture of ADHD through cross-sectional studies has been well-described (Sonuga-Barke et al., 2023). Studies examining genetic contributions to longitudinal changes in inattention and hyperactivity/impulsivity symptoms are limited but find that genetic factors contribute differentially to changes in symptom dimensions over time. For example, a study by Pingault et al. (2015) comparing identical and nonidentical twins found that the genetic influences on childhood ADHD severity are largely distinct from those driving symptom changes over time, although some genetic influences are shared. They also observed that genetics explained more variance in changes in hyperactivity/impulsivity symptoms (81%) relative to inattention (54%). However, classical twin studies do not allow us to pinpoint the roles of specific genes relating to baseline levels versus change over time.

ADHD, like many conditions, is highly polygenic. However, many of the key genes associated with ADHD diagnosis are enriched in the early pre/postnatal period (e.g., neurite outgrowth) or are active in later-maturing processes (e.g., maturation of dopaminergic/noradrenergic circuits). One possibility is that the genes relating to hyperactivity–impulsivity symptom change are linked to broad regulatory and developmental processes (e.g., maturation of cortico-striatal networks in adolescence), explaining the decline in these symptoms for many individuals over time. In contrast, baseline inattention symptoms may be more strongly related to genes linked to synaptogenesis and circuit refinement expressed in the early postnatal or childhood period, which have a relatively reduced impact on rate of change later in development. Thus, some genes may anchor early ADHD symptom levels, while other variants may be more instrumental in its later progression, particularly for hyperactivity/impulsivity. These hypotheses remain to be tested but provide a biologically informed framework for understanding differential developmental trajectories across symptom domains.

Process II. Effects of socioeconomic context

In parallel to genetic impacts on biological development, some environmental factors such as SES – which reflects access to material and nonmaterial resources – contribute independent variance to symptom trajectories. SES, often measured using indicators including income, educational attainment, and neighborhood quality, has a profound influence on development (Rakesh et al., 2025). Although isolating these effects is challenging due to confounding and gene–environment correlations, evidence supports their contribution to ADHD occurrence and progression (Faraone et al., 2024). This warrants consideration of SES and other environmental factors as predictors rather than confounds. For example, effects of family income are more relevant for ADHD prevalence in those without high genetic liability for ADHD (Rowland et al., 2018). This suggests that environmental influences such as low income – through, for example, increased stress or reduced support/cognitive stimulation – may play a greater role in ADHD when genetic risk is lower. Extending this logic across symptom domains, evidence showing that genetic contributions are greater for hyperactivity/impulsivity than inattention symptoms (Pingault et al., 2015) raises the possibility that change in inattention may be better explained by stress or support from environmental sources.

Process III. Environmental calibration of liability

Alongside independent influences of environmental factors, low SES may also moderate genetic influences on ADHD outcomes (Badini et al., 2024). Aligned with this idea, findings show that ADHD-associated genetic variants are often found in noncoding regulatory regions, which are more sensitive to environmental influences (Faraone et al., 2024). For hyperactivity/impulsivity, factors such as low stress and nutrition may promote gene expression for timely biological maturation of frontal brain regions beneficial for symptom decline. Meanwhile, reduced inattention may rely on genes linked to cognitive reserve or working memory capacity, and sufficient sleep, which can facilitate the development and refinement of compensatory mechanisms over time. Though speculative, these ideas can be examined in future work to establish how household or neighborhood environments may amplify or buffer liability resulting in symptom changes (with careful modelling considerations; Pingault et al., 2022). Although we focus on SES as an illustrative example, it is important to note that many environmental factors – such as childhood maltreatment, trauma, and institutionalization – contribute to ADHD symptomatology and potentially interact with genetic liability.

Developmental transitions as amplification windows

Development is also marked by significant transitions – including school transitions and puberty – where the balance of effects of genetics, environment, and their interaction on ADHD symptoms may be reweighted. For example, puberty involves developmentally triggered gene expression, which could appear in twin studies as new genetic effects (i.e., genetic innovation). It is also a time of increasing independence and mobility, resulting in greater exposure to neighborhood factors (e.g., safety)

© 2026 Association for Child and Adolescent Mental Health.

14697610, 2026, 5, Downloaded from https://acamh.onlinelibrary.wiley.com/doi/10.1111/jcpp.70150 by Stony Brook University Libraries, Wiley Online Library on [21/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

(Rakesh et al., 2025). These periods also coincide with rises in symptoms of other mental health conditions, increasing persistence risk (Caye et al., 2016). Careful study of these developmental transitions may reveal key genetic and environmental innovation effects linked to symptom changes. In addition, changes in symptom levels do not necessarily map onto changes in ADHD-related functional *impairment* (Sibley et al., 2022), and these periods of divergence represent a valuable opportunity to examine their potentially distinct associations with long-term outcomes.

Synthesis and future steps

It is undeniable that genetics have a strong influence over ADHD development, which warrants considerable empirical effort to unravel. Equally, environmental liability is the result of additive effects over multiple environmental factors, requiring substantial effort to unpack. Environmental effects have often been overlooked in prior longitudinal literature due to the high heritability of ADHD, and limited variance in twin studies associated with shared environmental effects (e.g., household and neighborhood factors) (Pingault et al., 2015). However, these effects require further research for several reasons. First, unlike underlying genetic liability, environmental factors can shift over time – some discretely (e.g., parental job loss, or trauma) or fluctuating (e.g., housing instability) – which have different impacts on symptom progression and cannot be captured by studies with relatively large intervals between time points (e.g., 2–4 years). Second, studies often assume a common average symptom trajectory across people, but 17%–64% of children with ADHD have a fluctuating symptom pattern over development potentially due to sensitivity to environmental variation (Norman et al., 2023; Sibley et al., 2022). Third, the environment is partly correlated with genetics through passive transmission of both genes and environments from parents, the way a child's genetic traits evoke responses from others, and children's active selection of compatible environments. Fourth, environmental effects most strongly linked to persistence risk have the potential to be mitigated through intervention. As one example, inclusive education settings increase odds of ADHD remission specifically in low SES contexts, providing one way forward to reducing disparities (Kim et al., 2019). Environmental factors, including both risk (low SES, trauma) and protective factors that may promote resilience (peer support, positive parenting), are a critical, yet underexplored, dimension that longitudinal research of ADHD symptoms has yet to fully address.

Synthesizing the ideas above, we endorse a framework that considers the effects of genetic factors, environmental factors, and their interplay in shaping inattention and hyperactivity/impulsivity symptom trajectories with the aim of mapping persistence pathways, while accounting for their shifting contributions over time. Methods such as latent growth curve and dynamic structural equation modelling, when paired with high-quality longitudinal data, can capture how specific genetic and environmental effects differentially influence symptoms over time. Such a framework would allow the testing of several hypotheses. For example, changes in hyperactivity/impulsivity symptoms may be more closely explained by genes linked to maturation of the cortex (e.g., fronto-striatal circuits) over adolescence and into adulthood, as well as interactions with environmental factors. By contrast, inattention symptoms may be more sensitive to conditions in the environment such as changes in stress or support. In addition, building on prior clustering studies, approaches such as latent class growth analysis or mixed effects modelling can be used to identify subgroups with distinct developmental trajectories and to characterize the genetic and environmental factors underlying these differences.

Summary

ADHD persistence represents more than a static continuation of early ADHD severity and genetic liability. It is the result of contributions from genes, environmental context, and their interaction over development, with specific testable mechanisms linked to hyperactivity/impulsivity versus inattention persistence. Applying unified models that dynamically partition variance from these different sources may improve both mechanistic insight and prediction. Further work aimed at identifying interventions and environmental supports that reduce the impact of genetic or environmental liability on ADHD persistence rates can reduce disparities and promote positive outcomes in the long term.

Phoebe Thomson and Divyangana Rakesh

Department of Neuroimaging, Institute of Psychology, Psychiatry & Neuroscience, King's College London, London, UK

E-mail: phoebe.thomson@kcl.ac.uk

Correspondence

Divyangana Rakesh and Phoebe Thomson, Department of Neuroimaging, Institute of Psychiatry, Psychology & Neuroscience, King's College London, London, UK; Email: divyangana.rakesh@kcl.ac.uk and phoebe.thomson@kcl.ac.uk

Acknowledgements

The authors have declared that they have no competing or potential conflicts of interest. DR is supported by a New Investigator Research Grant from the UKRI Medical Research Council (MR/Z506667/1).

© 2026 Association for Child and Adolescent Mental Health.

14697610, 2026, 5, Downloaded from https://acamh.onlinelibrary.wiley.com/doi/10.1111/jcpp.70150 by Stony Brook University Libraries, Wiley Online Library on [21/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

References

Agnew-Blais, J.C., Belsky, D.W., Caspi, A., Danese, A., Moffitt, T.E., Polanczyk, G.V., … & Lewis, C.M. (2021). Polygenic risk and the course of attention-deficit/hyperactivity disorder from childhood to young adulthood: Findings from a nationally representative cohort. *Journal of the American Academy of Child and Adolescent Psychiatry*, *60*, 1147–1156.

Badini, I., Ahmadzadeh, Y., Wechsler, D.L., Lyngstad, T.H., Rayner, C., Eilertsen, E.M., … & McAdams, T.A. (2024). Socioeconomic status and risk for child psychopathology: Exploring gene–environment interaction in the presence of gene–environment correlation using extended families in the Norwegian Mother, Father and Child Birth Cohort Study. *Journal of Child Psychology and Psychiatry*, *65*, 176–187.

Caye, A., Spadini, A.V., Karam, R.G., Grevet, E.H., Rovaris, D.L., Bau, C.H., … & Kieling, C. (2016). Predictors of persistence of ADHD into adulthood: A systematic review of the literature and meta-analysis. *European Child & Adolescent Psychiatry*, *25*, 1151–1159.

Faraone, S.V., Bellgrove, M.A., Brikell, I., Cortese, S., Hartman, C.A., Hollis, C., … & Rubia, K. (2024). Attention-deficit/hyperactivity disorder (primer). *Nature Reviews. Disease Primers*, *10*, 11.

Garner, A.A., O'Connor, B.C., Narad, M.E., Tamm, L., Simon, J., & Epstein, J.N. (2013). The relationship between ADHD symptom dimensions, clinical correlates, and functional impairments. *Journal of Developmental & Behavioral Pediatrics*, *34*, 469–477.

Kim, M., King, M.D., & Jennings, J. (2019). ADHD remission, inclusive special education, and socioeconomic disparities. *SSM – Population Health*, *8*, 100420.

Norman, L.J., Price, J., Ahn, K., Sudre, G., Sharp, W., & Shaw, P. (2023). Longitudinal trajectories of childhood and adolescent attention deficit hyperactivity disorder diagnoses in three cohorts. *EClinicalMedicine*, *60*, 102021.

Pingault, J.-B., Allegrini, A.G., Odigie, T., Frach, L., Baldwin, J.R., Rijsdijk, F., & Dudbridge, F. (2022). Research review: How to interpret associations between polygenic scores, environmental risks, and phenotypes. *Journal of Child Psychology and Psychiatry*, *63*, 1125–1139.

Pingault, J.-B., Viding, E., Galéra, C., Greven, C.U., Zheng, Y., Plomin, R., & Rijsdijk, F. (2015). Genetic and environmental influences on the developmental course of attention-deficit/hyperactivity disorder symptoms from childhood to adolescence. *JAMA Psychiatry*, *72*, 651–658.

Rakesh, D., Lee, P.A., Gaikwad, A., & McLaughlin, K.A. (2025). Annual research review: Associations of socioeconomic status with cognitive function, language ability, and academic achievement in youth: A systematic review of mechanisms and protective factors. *Journal of Child Psychology and Psychiatry*, *66*, 417–439.

Rowland, A.S., Skipper, B.J., Rabiner, D.L., Qeadan, F., Campbell, R.A., Naftel, A.J., & Umbach, D.M. (2018). Attention-Deficit/Hyperactivity Disorder (ADHD): Interaction between socioeconomic status and parental history of ADHD determines prevalence. *Journal of Child Psychology and Psychiatry*, *59*, 213–222.

Sibley, M.H., Arnold, L.E., Swanson, J.M., Hechtman, L.T., Kennedy, T.M., Owens, E., … & Rohde, L.A. (2022). Variable patterns of remission from ADHD in the multimodal treatment study of ADHD. *American Journal of Psychiatry*, *179*, 142–151.

Sonuga-Barke, E.J.S., Becker, S.P., Bölte, S., Castellanos, F.X., Franke, B., Newcorn, J.H., … & Simonoff, E. (2023). Annual Research Review: Perspectives on progress in ADHD science – From characterization to cause. *Journal of Child Psychology and Psychiatry*, *64*, 506–532.

Accepted for publication: 9 March 2026

© 2026 Association for Child and Adolescent Mental Health.

14697610, 2026, 5, Downloaded from https://acamh.onlinelibrary.wiley.com/doi/10.1111/jcpp.70150 by Stony Brook University Libraries, Wiley Online Library on [21/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License